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File Edit Selection View Go Run Terminal Help python fold
PYTHL...
> exel
adarsh.py
insurance.csv
insurance.py
model.pkl
rent
Rentpred.py
x_scaler.pkl
y_scaler.pkl

2 import pandas as pd
3 data=pd.read_csv("insurance.csv")
4 print(data.head())
5
6 from sklearn.preprocessing import LabelEncoder #labelencoder
7 label_encoder=LabelEncoder()
8 data['sex']=label_encoder.fit_transform(data['sex'])
9 data['smoker']=label_encoder.fit_transform(data['smoker'])
10 data['region']=label_encoder.fit_transform(data['region'])
11 x=data.drop(columns=["charges"])
12 y=(data['charges'])
13 print(x)
14
15 from sklearn.preprocessing import StandardScaler #standardscaler
16 x_scaler=StandardScaler()
17 y_scaler=StandardScaler()
18 x_scaled=x_scaler.fit_transform(x)
19 y_scaled=y_scaler.fit_transform(y)
20 #print(f"X Scaled:\n{x_scaled}")
21
22 from sklearn.model_selection import train_test_split #training the data
23 x_train,x_test,y_train,y_test=train_test_split(x_scaled,y_scaled,test_size=0.2,random_state=42)
24 #print(x_train.shape)
25
26 from sklearn.linear_model import LinearRegression #linear regression model
27 model=LinearRegression()
28 model.fit(x_train,y_train)
29 y_predict=model.predict(x_test)
30 print("coefficient :",model.coef_)
31 print("intercept:",model.intercept_)
32
33 from sklearn.metrics import mean_squared_error #regression metrics
34 error=mean_squared_error(y_test,y_predict)
35 print("mean square error :",error)
36
37 import numpy as np
38 rmse=np.sqrt(error)
39 print(f"Root Mean Squared Error: {rmse}")
40
41 import joblib #save the model and preprocessing objects using Joblib
42 joblib.dump(model, "model.pkl")
43 joblib.dump(x_scaler, 'x_scaler.pkl')
44 joblib.dump(y_scaler, 'y_scaler.pkl')
```

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PYTHON

exel

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PROBLEMSOUTPUTDEBUG CONSOLETERMINALPORTS

Python

PS C:\Users\QMEN\OneDrive\Desktop\python fold> & "C:/Program Files/Python314/python.exe" "c:/Users/QMEN/OneDrive/Desktop/python fold/adarsh.py"

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

	age	sex	bmi	children	smoker	region
0	19	0	27.900	0	1	3
1	18	1	33.770	1	0	2
2	28	1	33.000	3	0	2
3	33	1	22.705	0	0	1
4	32	1	28.880	0	0	1

... ..
1333 50 1 30.970 3 0 1
1334 18 0 31.920 0 0 0
1335 18 0 36.850 0 0 2
1336 21 0 25.800 0 0 3
1337 61 0 29.070 0 1 1

[1338 rows x 6 columns]

coefficient : [[2.98235092e-01 -7.76112153e-04 1.69088059e-01 4.23157876e-02
7.88313320e-01 -2.47512482e-02]]

intercept : [0.00371922]

mean square error : 0.229524876839686

Root Mean Squared Error: 0.47908754611207127

PS C:\Users\QMEN\OneDrive\Desktop\python fold>

Ln 3, Col 34Spaces: 4UTF-8CRLFPython3.14.5Python 3.14 (64-bit)


```
PS C:\Users\QMEN\OneDrive\Desktop\python fold> & "C:/Program Files/Python314/python.exe" "c:/Users/QMEN/OneDrive/Desktop/python fold/insurance.py"
Enter age: 48
Enter sex (0 = female, 1 = male): 0
Enter bmi: 27.8
Enter number of children: 2
Enter whether smoker or not (0 = non-smoker, 1 = smoker): 0
Enter region (0 = southwest, 1 = southeast, 2 = northwest, 3 = northeast): 0
   age  sex   bmi  children  smoker  region
0  48    0  27.8         2         0       0
medical insurance cost for a new individual.: [[3883.47859645]]
PS C:\Users\QMEN\OneDrive\Desktop\python fold>
```

Ln 10, Col 39 Spaces: 4 UTF-8 CRLF Plain Text